MOON-STONE SEMANTIC VAULT: REGULATORY COMPLIANCE MATRIX (PHASE 4)
Document ID: RCM-MS-PILOT-2026 | Target: DPO, InfoSec, and Regulators
Axiomatic Baseline: Monograph Chapters 6.5, 7.2, 7.4, 8, 15.4 and Appendix E
---

1. GDPR Art. 25 (Data Protection by Design & Default):
   - Mechanism: Architectural annihilation of F(x) at perimeter. Zero PII stored.
   - Proof: Axiom E.1: Proj_F(Sigma_pi(x)) = EmptySet. Surface form never hits disk.
   - Compliance: Full. Data is non-reconstructible by architectural design.

2. GDPR Art. 32 (Security of Processing):
   - Mechanism: Homophonic mapping with parameter rho and TEE isolation.
   - Proof: Corollary E.1: Absence of unique inverse R(g)=x.
   - Compliance: Complete. Cryptographic non-injectivity ensures zero reconstruction.

3. HIPAA Security Rule (164.312(a)(1) Access Control):
   - Mechanism: Model-as-Key Bound protocol (Tier 3). Requires cognitive topology M.
   - Proof: Chapter 7.4: Model Key Theorem. E_3 returns ONTOLOGICAL_MISMATCH on M'!=M.
   - Compliance: Institutional. Access strictly tied to certified neural topology.

4. SOC 2 CC6.1 (Logical Access Security):
   - Mechanism: Codebook Separation Axiom. C_research public, C_priv hardware-sealed.
   - Proof: Chapter 15.4: Federated Zero-Revelation Protocol. C_research cap C_priv = Empty.
   - Compliance: Verified. Zero data leakage between consortium and private tenants.

5. NIST IR 8397 (Confidential Computing):
   - Mechanism: Hardware TEE enclaves (SGX/SEV-SNP/Nitro) with hardware zeroization.
   - Proof: Chapter 7.2: Hardware Trust Boundary & anti-cold-boot RAM flush.
   - Compliance: Hardware-rooted. Intermediate states wiped post operator E_3.

Consortium Quorum: 3/4 Anchors Verified | Status: APPROVED FOR INSTITUTIONAL PILOT